

# Epistemic Injustice in Online Communities: Unpacking the Values of Knowledge Creation and Curation within CSCW Applications

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## ABSTRACT

Information flows are pervasive on the internet and often have a low entry barrier. From an epistemological perspective, information evolves into knowledge. For example, information about mental health on TikTok can act as actionable knowledge for someone seeking to improve their mental health. However, social computing has long known that people do not interact with knowledge cleanly, especially in digital environments. While knowledge curation is essential for targeting irrelevant, biased, or even harmful information, it is value-laden; in choosing how to present information, we undermine non-traditional information such as personal experiences. In this workshop, we will bring together researchers from academia, industry, and marginalized communities to discuss how current CSCW applications contribute to the systemic silencing, exclusion, or delegitimization of certain knowledge contributions (i.e., epistemic injustice). We will diagram our own mental models of how knowledge is created and curated and reflect on critical questions to orient the design of inclusive knowledge spaces online, particularly with topics that blend personal experience with factual information, such as mental health.

## CCS CONCEPTS

• Human-centered computing → HCI theory, concepts and models.

## KEYWORDS

peer production, online communities, epistemic justice

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## 1 INTRODUCTION

As the internet has become our main repository for information, it has simultaneously become a controversial venue for knowledge curation, creation, and dissemination. However, previous work in CSCW shows that individual values play a significant role in how knowledge-creation work is prioritized in peer-production [15] and how users theorize about their own online experience [7]. These conversations in HCI and CSCW are in harmony with feminist theories of situated knowledge: knowledge processes happen through a multitude of individual lenses and, therefore, are rarely “objective” [13]. While access to information and knowledge is progressing, it is unclear if the underlying epistemology—the theory behind what constitutes valid and justified belief—is similarly progressing. For example, Wikipedia has long been trying to close the gender edit gap to increase the encyclopedic coverage of the platform [1]. Yet, recent research suggests that this gap is in part due to fundamental platform policies that run antithetical to feminist epistemology rather than specific design features [18]. In social media settings, prior work finds there is a fine line between useful, personal mental health content and misinformation on TikTok [19]. These examples open up tricky questions about where to draw the line between useful community knowledge, biased information, and misinformation.

In this workshop, we will build upon work that highlights the nuance of legitimizing certain information as “knowledge” to call increased attention to harms in knowledge construction and curation. To scaffold this discussion, we will use Fricker’s framework of epistemic injustice [11] which has been cited in previous HCI work that critiques knowledge-based processes [22]. Specifically, we adopt her view that the systemic exclusion, silencing, or delegitimization of an individual’s contribution to the “knowledge canon” propagates the marginalization of voices that are typically already in the minority. In social computing systems, epistemic injustice can happen in the context of peer-production [18], online communities [9], citizen science [10], and various other platforms that rely on user contributions. Knowledge pathways online are often complex and involve numerous platforms that span a diversity of research areas. Moreover, individuals experience epistemic injustice differently, particularly when they have intersecting marginalized identities. Therefore, it is crucial we engage with a variety of perspectives in a well-facilitated environment. We propose this workshop as an organizing space where we can unpack the relationships between platforms, policies, and values so we can better understand potential causes of epistemic injustice in the systems that we research.

## 2 KEY THEMES

Our workshop will focus on three key themes around knowledge construction via online settings: 1) creating a research agenda to understand the various pathways users take to create knowledge, 2) distinguishing the platforms, policies, and tensions that contribute to epistemic injustice along these pathways, and 3) identifying useful collaboration opportunities across contexts to fight epistemic injustice in online spaces.

### 2.1 Catalyzing a Research Agenda on User Pathways to Create Knowledge

Several critiques of popular conceptualizations of knowledge explore how a user journey may incorporate multiple contexts [8, 14]. Below we provide some examples of such journeys. These will additionally serve as prompts for participants to construct a research agenda and gain a more comprehensive understanding of potential research directions in this space.

- **Common and Factual Knowledge:** Common knowledge is factual knowledge that is publicly known or accessible to everyone [2]. Historically, this knowledge existed in libraries. However, as information has shifted online, the boundaries of common knowledge have become blurry. Wikipedia is the most notable curator of online knowledge, designed with the intention of being an accessible encyclopedia [16]. News websites also originated as factual institutions. However, as news sites move online, they are trying to promote substantive discussion in a factual space via online comment sections [17]. Recent work has found that clinical information is prevalent on social media websites, such as TikTok, and can sometimes guide users on their own mental health journeys [19]. We invite conversations about how factual knowledge can occupy the same space as personal experience in online settings.
- **Shared Knowledge and Norms:** Epistemists often note that knowledge is a collective resource; shared knowledge is a fundamental component in creating a sense of community [20, 24]. Moreover, the implicit norms of a community are an integral piece of knowledge for participation. We ask: what are the values and tensions that arise during the process of creating shared knowledge in online communities? In online health communities, unmoderated norms can contribute to deviant behaviors [4]. However, recent work has highlighted how the LGBTQ+ community values agency to create their own norms outside of platform influence [9].
- **Anecdotal evidence and personal experience:** Recent work has shown that anecdotal evidence can be compelling in online spaces. For example, citizen science communities can take anecdotal data and incorporate it into a robust experiment workflow [21]. In social media settings, personal experience mediates feelings of peer support. Moreover, it can catalyze a knowledge journey about one's mental health status [19]. However, an over-reliance on lived experience can create rampant misinformation, particularly in online health communities [26]. Given that anecdotal evidence contributes to user knowledge, we encourage researchers to discuss how to appropriately legitimize personal experience

in digital spaces. Where does this tacit knowledge belong online?

### 2.2 Contributors to Epistemic Injustice

In online settings, there is a constant debate about what constitutes "contribution." This workshop aims to promote discussion around how these tensions can lead to epistemic injustice in digital spaces. We invite researchers to interrogate how users in knowledge pathways are systemically excluded, silenced, or delegitimized. Specifically, we ask: how do certain platforms, policies, and institutions contribute to epistemic injustice? We will provide case studies from workshop submissions. We list two case studies below as inspiration:

**2.2.1 Case study 1: Gender Dynamics on Wikipedia.** The gender edit gap has been a prominent problem on Wikipedia for nearly a decade [1, 5]. In popular media, this has led to wide controversies such as the exclusion of Kate Middleton's dress from Wikipedia [3] and the silencing of experience surrounding the GamerGate harassment campaign [23]. Menking and Rosenberg [18] recently noted that the gender edit gap on Wikipedia is an epistemic problem rather than a design one. They note that the fundamental policies of Wikipedia are not aligned with feminist work on how to create inclusive online knowledge spaces.

**2.2.2 Case study 2: LGBTQ+ Knowledge and Content Moderation on Social Media.** Members of the LGBTQ+ community struggle to get useful and actionable information from traditional sources. Therefore, online communities and social media often serve as crucial knowledge sources [25]. Despite the importance of these online spaces, LGBTQ+ folks face an outsized burden of content moderation when systems over filter shared vocabulary [12]. Effectively, these policies on social media websites, such as TikTok, put a rigid constraint on what is "acceptable" knowledge and leave folks with nothing to turn to. Traditional online sources are not designed for marginalized communities, but non-traditional knowledge spaces, such as social media, are moderated such that these communities cannot exist in full.

### 2.3 Roadmapping Directions towards Fighting Epistemic Injustice in Online Spaces

Building upon participants' reflections on tensions and potential harms, we invite discussion on creating a roadmap for combating sources of epistemic injustice in digital settings. Given the long history of CSCW's focus on knowledge, value-sensitive design, and justice, we are particularly interested in exploring how future research directions can promote designs and policies that raise voices rather than exclude them. Inspired by feminist epistemology [13, 18], below are a few possible guiding questions:

- **Is online knowledge curation value-laden?** In alignment with situated knowledge theory [13], can we unpack our notions of "objectivity" in a way that accounts for value-ladenness? Can we go beyond concepts of bias and, rather, deeply articulate the values we bring to the table in deciding how knowledge is created and curated?
- **How can we properly hold space for personal experience when designing online platforms and policies?**

What online venues are appropriate for personal experience versus not? In light of new platforms, such as TikTok, that display personal experience in the same venue as clinical information [19], how can we separate anecdotes without delegitimizing them?

- **How can we design for the systemic factors contributing to epistemic injustice?** What can researchers do to combat the deeply entrenched systems that make online spaces ripe for epistemic injustice?

### 3 WORKSHOP LOGISTICS

We plan to host a one-day hybrid workshop including a moderated panel discussion, lightning talks, small group activities, and a debriefing to foster future collaboration. To accommodate time zone differences for virtual participants, we have divided our workshop into two half-day sessions. Because the two sessions will not overlap in content, we welcome and encourage participants to join both sessions.

#### 3.1 Description of Activities

**3.1.1 Lightning talks.** We will have two blocks of lightning talks and each speaker will have five minutes to present their extended abstracts. From past experience, we find that having lightning talks helps participants better understand each others' backgrounds and stimulates more constructive discussion. Speakers will be able to give their presentations in-person or virtually through Zoom. Each talk will be followed by a Q&A session that blends questions from virtual and in-person participants. An organizer will be equipped to facilitate virtual discussion as needed.

**3.1.2 Moderated Panel.** We will recruit four to six panelists to explicate the nuances of knowledge in digital spaces, particularly with respect to marginalized communities, which provides important context for our workshop's themes. We will specifically recruit panelists who are members, advocates, and allies of the marginalized voices that are common victims of epistemic injustice. We hope our panel diversity serves as an in-process way of enacting our overall goal of legitimizing contributions that are commonly silenced or excluded. To support a hybrid event, participants will be able to ask questions through a multitude of avenues: (1) through an anonymous form ahead of the workshop, (2) in-person during the panel, and (3) through a virtual chat during the panel. Organizers will monitor each venue to ensure there is proportional representation from in-person and virtual participants.

**3.1.3 Small group activities.** We will have two small group activity sessions. To create a seamless hybrid experience, small groups will be divided by modality. In-person groups will be given post-it notes, pens, and blank space while virtual groups will be given pre-arranged Miro boards that can be projected in the workshop room. An organizer will be responsible for ensuring that hybrid participants are given ample opportunity to contribute to any larger discussions.

**Session 1: Visual Elicitation of Knowledge Pathways.** In the first session, we will focus on reaching a shared understanding of the various knowledge pathways that exist in digital settings (Theme 1). The output of this activity will be diagrams that map the

user journey through a multitude of platforms such as Wikipedia, social media, and peer-support communities to shape their knowledge on a topic. These diagrams will serve as a foundation for the second small group activity. Visual elicitation exercises are commonly used in qualitative research to help interviewees communicate abstract ideas that can be difficult to discuss [6]. To limit the scope of this activity, each small group will focus on a specific user path.

#### Session 2: Identifying contributors to Epistemic Injustice.

Once we have established some of the platforms that shape knowledge in digital spaces in Session 1, we plan to identify contributors to epistemic injustice. We ask: are there areas of these knowledge pathways that are ripe for epistemic injustice? To scaffold this discussion, we will introduce the case studies from submissions and ask how these play out across the entire ecosystem rather than on any single platform.

**3.1.4 Debriefing.** We will have a short debriefing session at the end of the workshop summarizing key inputs from participants and mapping out the main conclusions from activities. We will also provide the space for participants to brainstorm about establishing collaborations with different disciplines and groups moving forward to combat epistemic injustice in online spaces.

#### 3.2 Schedule

As stated above, we plan to have two hybrid half-day sessions with distinct but complementary discussions. All times are in North America Central Time.

##### First half-day session:

- 9:30am - 10am: Intro to Workshop
- 10:10am - 11:00am: Lightning talks and Q&A
- 11:10am - 12:00pm: Visual Elicitation activity

##### Second half-day session:

- 1:00pm - 2:00pm: Moderated panel
- 2:10pm - 3pm: Lightning talks and Q&A
- 3:10pm - 3:40pm: Small group activity on values
- 3:40pm - 4pm: Debrief and roadmapping discussion

Materials needed for the workshop include post-it notes, markers, and large surfaces that post-it notes can be adhered to (e.g., walls or whiteboards). In addition, we would appreciate the A/V technology to support a hybrid setting, including a projector, web camera, speaker, and microphones.

#### 3.3 Participant Recruitment

We expect a maximum of 40 participants for this workshop. We will use social media and mailing lists to invite academic researchers to submit extended abstracts. We will also invite experienced members of the communities being discussed, e.g. Wikipedians, Reddit moderators, and citizen scientists, to participate in our panel and dialogue with researchers. Moreover, we plan on inviting members of special interest groups such as Wikimedia LGBT+<sup>1</sup> and Women in CSCW Facebook group. We are in the process of securing funding to support the attendance for their participation in our workshop.

<sup>1</sup>[https://meta.wikimedia.org/wiki/Wikimedia\\_LGBT%2B/Portal](https://meta.wikimedia.org/wiki/Wikimedia_LGBT%2B/Portal)

**Participant Submissions and Review Process.** Individuals interested in participating will be asked to submit an extended abstract for a 5-minute lightning talk they will give as part of the workshop activities. Submissions should be 2-4 pages long and outline the participants' research interests and how they relate to the key themes of our workshop. Accepted abstracts will be used to create activity subgroups and adjust any activities appropriately.

The organizing team will review submissions and seek external reviewers if necessary. We plan to have two to three reviewers for each submission. Reviewer comments will be shared with the authors. We welcome early-stage analytical work, case studies, essays, and design fictions.

### 3.4 Workshop Modality, Accessibility, and Accommodation

The workshop will be held in a hybrid format. We believe that the in-person option will promote lively discussion and activities. Meanwhile, the hybrid option will ensure we are making our workshop more inclusive and encourage a wider array of participants to join the discussion. We will provide multiple opportunities for asynchronous engagement if individuals cannot attend synchronously. With the speakers' permission, we plan to record and disseminate videos of the lightning talks. We will share an accessible version of the diagram outputs from our activities publicly on our website. Additionally, the community will be able to engage with submissions before, during, and after the workshop via virtual "pinboards" created with Miro.

### 3.5 Workshop Goals

This workshop aims to bring together researchers in the community working on a diverse range of topics related to knowledge construction in online settings. We hope to involve an interdisciplinary group of researchers who may not otherwise have the opportunity to engage in an extended dialogue about epistemic injustice. The workshop activities are designed to facilitate the sharing of research ideas across specialties; highlight common threads between adjacent areas, such as peer-production and online communities; and understand how our research can mitigate epistemic injustice in online spaces.

### 3.6 Post-Workshop Plans

We will explicate our workshop outcomes in the format of a blog post or submission to the Communications of ACM. We hope that, by distributing the outcomes of our workshop more broadly, we can engage with more members of the CSCW community in the discourse on epistemic injustice in digital spaces. We will also create a mailing list for participants to facilitate post-workshop communications.

## 4 ORGANIZERS

Our organizing team consists of researchers from a diversity of disciplines (HCI, computational social science, information systems) across five institutions, who collectively contribute expansive research expertise on various knowledge-seeking behaviors online.

**Leah Ajmani** is a PhD student in Computer Science at the University of Minnesota. She studies policies and ethics in mental health settings and online communities. Recently, she has been working on how Wikipedia policies exacerbate pre-existing barriers to contribution.

**Mo Houtti** is a PhD student in Computer Science at the University of Minnesota. He studies Wikipedia's values and how those values interact with article prioritization on the platform.

**Jasmine Foriest** is a PhD student in human-centered computing at Georgia Institute of Technology. She studies digital muting of non-dominant groups, in health and justice settings, and the implications for knowledge creation and social outcomes.

**Nicholas Vincent** is a postdoctoral researcher at the University of California, Davis. His work focuses on studying the dependence of modern computing technologies, including peer-production systems that rely on human contribution, with the goal of mitigating the negative impacts of these technologies.

**Michael Ann Devito** is an Assistant Professor of Computer Science and Communication Studies at Northeastern University. She specializes in human-computer interaction on social platforms, particularly with members of the LGBTQ+ community.

**Isaac Johnson** is a research scientist at the Wikimedia Foundation. His work focuses on the policies and interactions that facilitate crowd-constructed knowledge.

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